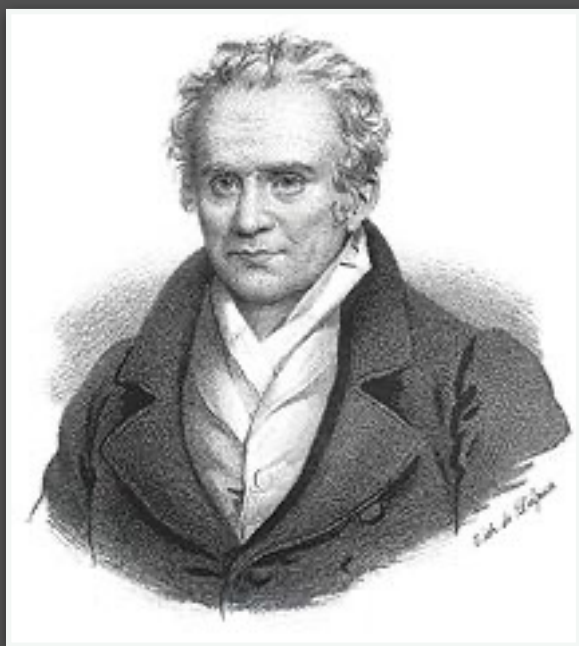


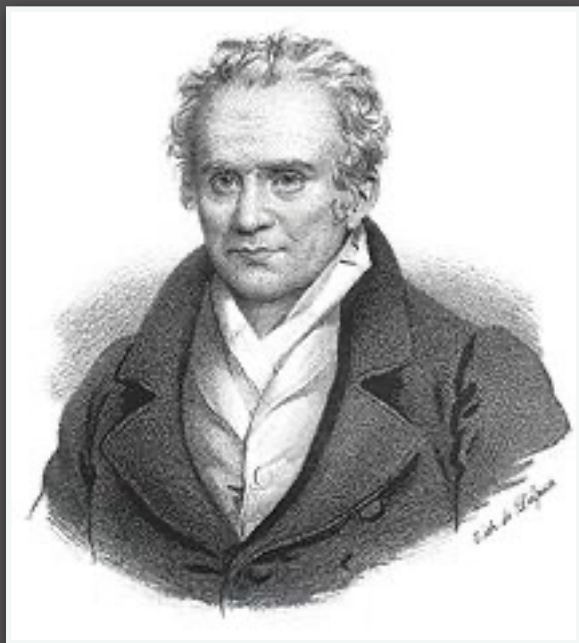
Last Time



Monge's Problem:

(Matching/permutation version)

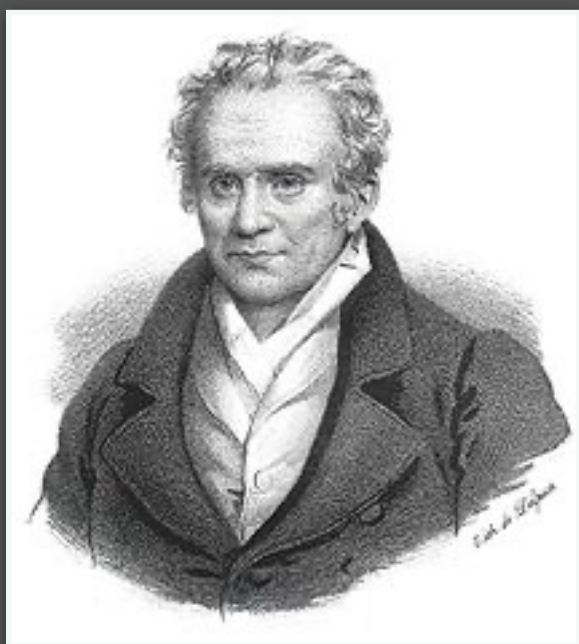
$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^n C_{i, \sigma(i)}$$



Monge's Problem:

(Discrete measure version)

$$\min_{T_{\#}\alpha = \beta} \frac{1}{n} \sum_{i=1}^n c(x_i, T(x_i))$$



Monge's Problem:

(Continuous / general version)

$$\min_{\beta = T_{\#}\alpha} \int_X c(x, T(x)) d\alpha(x)$$

Theorem [Brenier, 1991]:

Assumptions: $c(x, y) = \|x - y\|^2$, α has a density $\rho_\alpha = \frac{d\alpha}{dx}$

Then:

- Monge problem has a **unique** solution T
- T is the gradient of a convex function

OT between Gaussians has closed form

$$\alpha = \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$$

$$\beta = \mathcal{N}(\mu_\beta, \Sigma_\beta)$$



$$T(x) = \mu_\beta + A(x - \mu_\alpha); \quad A = \Sigma_\alpha^{-\frac{1}{2}} (\Sigma_\alpha^{\frac{1}{2}} \Sigma_\beta \Sigma_\alpha^{\frac{1}{2}})^{\frac{1}{2}} \Sigma_\alpha^{-\frac{1}{2}}$$

Also Last Time